Task W11: Advanced Trees

Root Canal

A note on how this assignment was developed

This assignment was drafted with AI help (mostly for brainstorming new problems to solve and formatting with LaTeX). It should be accurate, but something might still be wrong. Because this is a new assignment, you are also helping me debug it, and I will take that into consideration when I review your work. That said, helping me find rough edges is not a substitute for doing the assignment; each problem below is solvable, but you may need to make assumptions in your attempt. If you find an error or something unclear, please tell me; that helps later versions, though finding errors is not the point of this assignment.

Overview

This task has three problems, one for each kind of balanced tree work we do in lecture. Problem 1 builds trees by insertion: the same list of values goes into a 2-3 tree, a 2-3-4 tree with preemptive splitting, and a 2-3-4 tree without preemptive splitting. Problem 2 deletes values from a 2-3 tree, and Problem 3 deletes values from a 2-3-4 tree.

Follow the algorithms from our book and lecture, not a variation you find online. Balanced trees have several published variations, and they build different trees from the same values, so an answer that follows a different version will not match.

Diagram Submission Instructions

Every problem asks for a diagram. You may choose one of the following options.

- **Hand-drawn on paper or a whiteboard.** Scan or take clear, well-lit photos. Images must be easy to read and clearly labeled with problem numbers. For this task in particular, this is the recommended option: trees are much faster to sketch by hand than to build in software.
- **Hand-drawn using an app** (for example Vittle Pro or Adobe Fresco). Save as an image file (PNG, JPG) or PDF. This is a convenient option but may be harder without a stylus.
- **Created with diagramming software** (for example Draw.io, LibreOffice Draw, or Excalidraw). Save in the tool's native file format; do not export as an image or PDF.
- **Other tools.** If you would like to use a different diagramming application, ask for approval first. Note that I have limited installation access on my office computer.

Include your name and the problem number, and the part letter for Problem 1, on every diagram.

If you find something that needs to be debugged

If, while working through this assignment, you discover something in the problem statements that needs to be debugged, submit a separate document describing what you found. This is separate from, and in addition to, your answers.

Where to submit

Put everything for this task (your diagrams and any debugging document) in a folder named `taskw11` in your GitHub repository, and commit it.

Problem 1: Inserting into Balanced Trees

Insert the following values into each tree below, reading from left to right. Each tree starts empty.

26 73 97 22 58 29 94 68 93 79

(a) A 2-3 tree.

(b) A 2-3-4 tree with preemptive splitting.

(c) A 2-3-4 tree without preemptive splitting.

You only need to submit a drawing of the final tree for each part, after all the inserts.

Problem 2: Deleting from a 2-3 Tree

Starting with the 2-3 tree in Figure 1, delete the following values, reading from left to right.

48 71 41 47 21

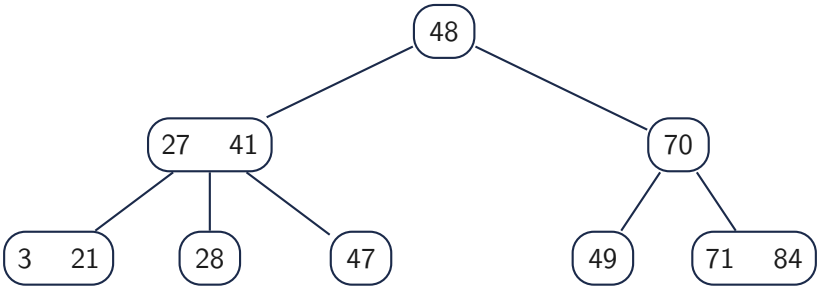


Figure 1: The 2-3 tree for Problem 2. The root holds 48. Its left child holds 27 and 41 and has three leaves: 3 21, 28, and 47. Its right child holds 70 and has two leaves: 49, and 71 84.

You only need to submit a drawing of the final tree, after all the deletes.

Problem 3: Deleting from a 2-3-4 Tree

Starting with the 2-3-4 tree in Figure 2, delete the following values, reading from left to right.

10 83 34 32 50

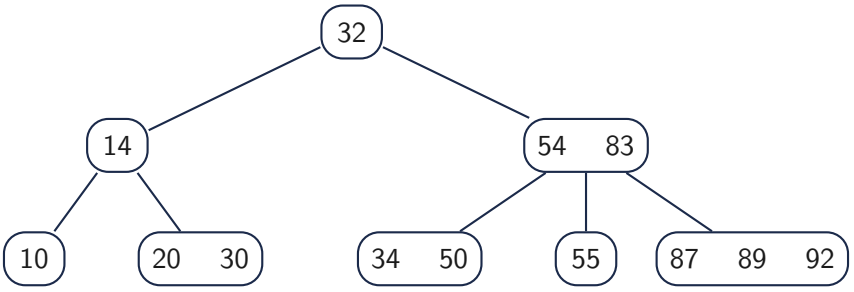


Figure 2: The 2-3-4 tree for Problem 3. The root holds 32. Its left child holds 14 and has two leaves: 10, and 20 30. Its right child holds 54 and 83 and has three leaves: 34 50, 55, and 87 89 92.

You only need to submit a drawing of the final tree, after all the deletes.

What to Submit

- **Problem 1:** three diagrams, the final tree for each of parts (a), (b), and (c).
- **Problem 2:** one diagram, the final 2-3 tree after all five deletes.
- **Problem 3:** one diagram, the final 2-3-4 tree after all five deletes.
- **Optional:** a separate document describing anything in the problem statements that needs to be debugged.

Everything above goes in the `taskw11` folder in your GitHub repository.